PredictWise AI

The simple guide: what is inside, and how to show it

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What it is

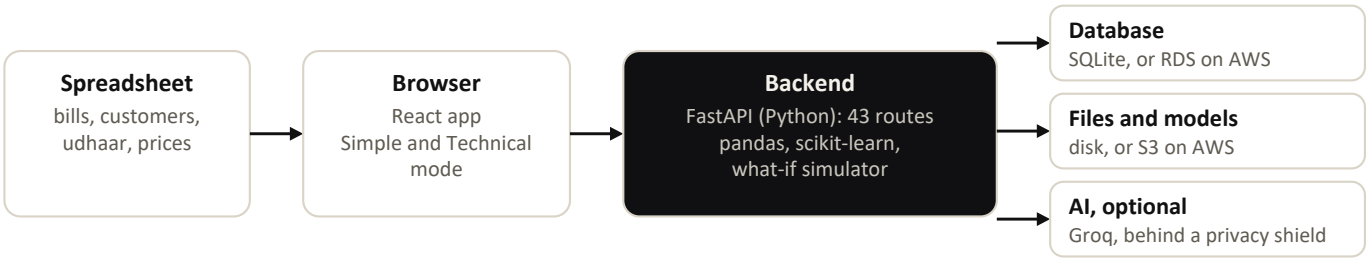
PredictWise AI is a web app for small shops. The shopkeeper uploads a spreadsheet of bills. The app then answers questions about it in plain words, in English, Hindi or Hinglish. **Every number is calculated from the file by code, so it can be checked. Nothing is guessed.**

Why not just ChatGPT? A general chatbot cannot see your file and guesses the numbers. PredictWise calculates them from your own file, shows the working, keeps customer names on your computer, and most of it works with no internet.

Two modes, one engine

Simple mode	For the shopkeeper. Four tabs: **Chat**, **Dashboard**, **Look up**, **What if**. A toolkit of ready answers sits in the left column.
Technical mode	For the analyst and the examiner. Six stages: **Data, Clean, Explore, Model, Predict, Audit**.
Switching	Press **Simple** or **Technical** in the header. The same file and the same engine sit behind both.

How it fits together



On AWS the same code runs behind a load balancer on several servers. One setting switches the app from local to cloud.

The sample files (one click on the start screen)

FILE	WHAT IT IS	USE IT FOR
Sharma General Store	16 months of bills: 39,621 lines, 84 customers, 20 products, udhaar, last year's Diwali.	**Simple mode.** Every tool lights up.
Udhaar khata	A credit ledger: 600 customers, 70% repaid on time. Has blanks, repeats and two absurd amounts.	**Technical mode.** Clean, train, predict, audit.
Provision store, Kirana store, Sweet shop, Customers	Four smaller samples from earlier days.	Quick extras. Customers is a good yes/no example.

Before you start (two minutes)

1. Double-click **start.bat**. It starts the backend and the website and opens **localhost:5173**.
2. The left column must say **AI assistant · on (Groq)**. If it says off, restart the backend window.
3. Open the Sharma sample once, so the first briefing is ready before the audience arrives.
4. The free AI answers about two or three open questions a minute. The dashboard, look-up, What if and toolkit never use it.
5. Keep the Udhaar khata sample one click away for Technical mode.

Simple mode: what is inside and how to show it

Load **Sharma General Store** from the start screen. Rows marked with a tab name are in the tab bar above the chat.

PART	WHAT IT DOES	HOW TO SHOW IT
Briefing	Appears by itself with the file: period, total, trend with forecast, top items, unusual rows, data health.	Open the sample. Nothing to click.
Chat	Ask in plain words. Each answer is labelled: calculated from your file, AI answer, or model prediction.	Type *Who owes me money?*
AI answers	Open questions go to the AI. It writes one query, the code runs it, and every figure is checked.	Ask *Which area buys the most atta?* then open **How I worked this out**.
Dashboard (tab)	The shop at a glance: trend, sales by item, this week's actions, busy days, udhaar, profit, festival stock, customer groups, next week's order, unusual entries.	Click **Dashboard**. Click **Ask** on any tile.
Look up (tab)	One customer or product: spend, rank, last visit, udhaar, favourites, a ready WhatsApp message.	**Look up**, then *Dinesh Sharma*, then *Cold Drink 2L*.
What if (tab)	Rehearses a decision: the next 30 days replayed 2,000 times from the shop's own last 12 weeks. Six levers: close a day, change prices, run an offer, drop a product, stop udhaar, call back quiet customers.	**What if**: switch on *Close one day a week* and *Change prices*.
Shop toolkit	Left column. Profit, udhaar, slow stock, festival stock, bought together, busy days, order plan, win-back list, WhatsApp texts.	Click *Who owes me money?*, then *Reminder message for udhaar*, then **Send on WhatsApp**.
Order plan	Next week's order per product. It replays four methods on past weeks, picks the best, and checks itself on weeks held back.	Click *How much should I order?*
Hindi and voice	Answers in English, Hindi or Hinglish. Speak a question, listen to an answer.	Press the Hindi button and ask again. Click the microphone.
Private mode	Nothing is sent to any AI service. Everything else still works.	Switch it on, ask an open question: it says what it can do offline.
Question library	Ready questions in groups, built from this file's columns.	Click the book icon beside the question box.
Predict in chat	Customers sample: a mini form, then the answer with a trust note, track record, the reasons, and what would help most.	Open **Customers**, click *Will this happen again?*, then **Get my answer**.
Actions	Share today's summary, compare with an older file, download cleaned data, copy, listen, share on WhatsApp.	Left column, under **Actions**.

Technical mode: stage by stage

Use **Udhaar khata**. Load it in Simple mode, then press **Technical**. The pipeline is the rail on the left. Stages unlock in order.

STAGE	WHAT IT DOES	SAY	SHOW IT
01 Data	Lists the files kept in the database. Each upload is a row with its stored path, row count and columns.	*Every upload is a row in the database.*	Point at the file list and the badge **Local · SQLite · Disk**.
02 Clean	Fills gaps, drops repeated rows, caps extreme values. Every step is logged in the **cleaning_logs** table.	*Preprocessing is logged, not hidden.*	Set **Extreme values** to *Cap using the IQR rule*, press **Clean dataset**, read **What changed**: 606 to 600 rows, 18 ages and 12 bills filled, outliers capped.
03 Explore	Shape, missing values, a histogram per column, a correlation grid.	*The analysis an analyst does by hand, in one click.*	Press **Continue to exploration**. Point at the grid: blue is positive, red is negative.
04 Model	Trains four algorithms, ranks them by 5-fold cross-validation, scores the winner once on rows held back, shows what drives the answer, saves the model.	*Choose by cross-validation, report on rows the model never saw.*	**Column to predict** is already *repaid_on_time*. Press **Train models**. Read the three panels: Selected model, Model leaderboard, What drives the prediction.
05 Predict	A prediction with confidence, a trust note, a track record, the reasons (Shapley) and a what-if curve. Every prediction is logged.	*It knows what it doesn't know.*	Type the safe row below and press **Predict**. Then set udhaar amount to 99999: **Outside the data**.
06 Audit	Fairness by group under the four-fifths rule, live monitoring of inputs, printable reports.	*A screening tool, not a legal audit.*	**Group column** Area, **Run audit**. Then Gender. Open **Plain-language report**.

Numbers to remember (Udhaar khata)

- 600 customers after cleaning. **70% repaid on time**, so always guessing “repaid” would score about 70%.
- Leaderboard, 5-fold accuracy: **Logistic Regression 0.81**, Random Forest 0.80, Gradient Boosting 0.78, Decision Tree 0.71. The simple model wins: the app picks by score, not by assuming that complex is better.
- On the 120 rows held back: accuracy **0.80**, F1 0.80, ROC AUC **0.90**.
- Biggest driver by far: **late_before** (earlier late payments). Then udhaar amount, days since last visit, area.
- Fairness by area: Tilak Nagar gets “repaid” **40%** of the time against **88%** for Jawahar Nagar. Ratio 0.46, below the 0.80 guideline. The same gap is in the real data. By gender the ratio is 0.86: no large gap.

Rows to type on the Predict stage

Safe: area Jawahar Nagar, gender Female, age 38, months as customer 36, visits per month 12, avg bill 420, udhaar amount 1500, previous udhaar 3, late before 0, days since last visit 4. *Result: repaid, about 99%, familiar input.*

Risky: change area to Tilak Nagar, udhaar amount to 99999, late before to 3, days since last visit to 90, visits per month to 3. *Result: Outside the data, with the reason for each input.*

How the AI is kept honest

1. **The AI never does the maths.** It writes one pandas query. The code runs it in a restricted sandbox and hands back the number.
2. **Every figure is checked.** Each number in the answer must appear in what the query returned. The check is shown under the answer.
3. **Customer data stays home.** Names, phones and emails become codes before anything is sent, and come back in the answer. Private mode sends nothing.
4. **It degrades, it does not fail.** With no key or no internet, the toolkit, forecasts, groups and simple questions still work.

How we know the numbers are right. 116 automated tests (`python -m pytest tests -q`, about 2.5 minutes). A 30-question accuracy suite (`python evals.py`) compares the app's answers with independent pandas calculations: **30 of 30 match**. Both run on GitHub Actions.

The technology, and how to show each piece

TECHNOLOGY	WHAT IT DOES HERE	HOW TO SHOW IT
React, Vite, Tailwind	All the screens, both modes, the dark and light theme.	Open **localhost:5173**. Click the sun or moon button.
FastAPI (Python)	The backend: 43 routes for upload, clean, train, predict, audit, chat.	Open **localhost:8000/docs**: the live list of every route.
pandas, NumPy	Every calculation in the toolkit, dashboard and look-up. Also the what-if simulator.	Open **How I worked this out** under an AI answer.
scikit-learn, joblib	Cleaning pipelines, four algorithms, saved models.	Technical, Model. Saved models sit in **backend/artifacts**.
SQLAlchemy, SQLite or PostgreSQL	Tables: users, datasets, cleaning_logs, experiments, ml_models, predictions, fairness_audits.	Header badge, or run the one-liner under this table.
Groq (the AI)	Open-ended chat, and Hindi wording.	The AI badge in the left column. Ask a question. Switch on Private mode.
Docker, GitHub Actions	One container for the whole app. Tests and builds on every push.	Show the **Dockerfile** and the Actions tab.
pytest, evals.py	The 116 tests and the 30-question accuracy check.	Run the two commands in a terminal.

Show the database tables:

```
python -c "import sqlite3; print(sqlite3.connect('backend/predictwise.db').execute('select id,file_name,row_count from datasets').fetchall())"
```

The cloud, in plain words

- **One command builds it:** `infra\deploy.ps1` creates four CloudFormation stacks: network, data, app, monitoring. `infra\teardown.ps1` removes them.
- **Request path:** browser, load balancer (two availability zones), EC2 Auto Scaling group (1 to 4 servers running the Docker image), then RDS PostgreSQL and an S3 bucket.
- **Events and watching:** when a CSV lands in S3, a Lambda profiles it. CloudWatch has a dashboard and 7 alarms. Secrets Manager holds the database password and the API key.
- **Cost:** lean mode is about $0.08 an hour (one server, no NAT gateway). Tear it down after the demo.
- **Show it:** the header badge reads **Live on AWS · S3 · RDS PostgreSQL**. `/health` shows which server answered. Then the S3, RDS and CloudWatch pages in the console.

Run sheet: about twelve minutes

WHEN	DO	SAY
Simple, 5 min	Open Sharma. **Dashboard**. **Look up** Dinesh Sharma. **What if**: close a day, raise prices 5%. Ask the atta question, open the working.	“Every tile is calculated from the shop's own bills.”
Technical, 5 min	Udhaar khata, **Technical**. Clean with capping. Explore. **Train models**. Predict a safe row, then udhaar 99999. Audit by area, then gender.	“Choose by cross-validation, report on unseen rows.”
Cloud, 2 min	Header badge. `/health`. S3, RDS, CloudWatch in the console. If AWS is not deployed, show the badge **Local · SQLite · Disk** and the four templates in **infra**.	“The same code, one setting different.”
Short on time	Dashboard, What if, one AI question, Train models, Predict.	
If something breaks	Switch on **Private mode** (all but open AI still works) and carry on with the Technical stages, which need no AI.	

Questions you may be asked

Why not ChatGPT?	It is great for writing. Here the job is records to decisions: numbers come from code run on your file, the working is shown, names stay on your computer.
What happens to my data?	The file stays on the computer running the backend. Names, phones and emails become codes before anything goes to the AI. Private mode sends nothing.
Why cross-validation and a held-out set?	Cross-validation picks the model fairly. The held-out rows give an honest score on data it never saw. Using one set for both flatters the model.
Why does the held-out score differ a little from the cross-validation score?	The held-out set is only 120 rows, so it moves a few points. Cross-validation averages five folds and is the steadier number.
Why did Logistic Regression win?	The risks here add up almost in a straight line, so a simple model is enough. The app picks by score, not by assuming that complex is better.
Why are some importances negative?	Shuffling that column happened to make the held-back score a little better, so it carries no real signal. The order of the bars is what matters.
Is the What if simulator machine learning?	No. It replays the shop's own days. What a file cannot say (price reaction, offer lift, customers returning) is shown as assumptions you can change.
What are the limits?	One shop per install, no login yet, tested on sample data not a field trial, festival dates are approximate, and Hindi answers to open questions need the AI key.

Words explained

Udhaar	credit given to a customer, to be paid later.
Cross-validation	train and test on different slices five times and average, to pick a model fairly.
Held-out rows	rows kept aside and never used to choose a model: the honest exam.
Leakage	test information sneaking into training, which inflates the score.
Permutation importance	shuffle one column; the drop in score shows how much the model relied on it.
Shapley values	a fair split of one prediction among its inputs; they add up exactly to the answer minus the average.
ROC AUC	how well the model ranks likely repayers above unlikely ones: 0.5 is chance, 1.0 is perfect.
Four-fifths rule	a group's rate should be at least 80% of the best group's rate.